

COURSE OVERVIEW

1. **RTAC Architecture:** The SEL RTAC — available as the SEL-3505, SEL-3530, and SEL-3555 — is a protocol gateway, data concentrator, and local logic controller in a single DIN-rail device. It sits between field devices speaking DNP3, Modbus, and IEC 61850, and the SCADA master that speaks DNP3 or IEC 61850. Hardware variants, I/O module selection, communication ports, and the physical commissioning steps before software is involved.
2. **ACCELERATOR Architect:** Every RTAC project is a database. Tags represent individual data points. Aliases map protocol point addresses to tag names. Revision control logs every configuration change with a timestamp and operator ID. The project workflow from a blank project to first live data.
3. **DNP3 Configuration:** The RTAC communicates upstream to the SCADA master as a DNP3 outstation, and downstream to field devices as a DNP3 master. Point database mapping, event class assignment, unsolicited reporting intervals, and the DNP3 parameters that determine latency from a field change to a SCADA display update.
4. **Modbus Gateway:** The RTAC exposes a Modbus TCP server that maps DNP3 analog and binary points to holding registers and coils. Legacy SCADA systems that speak only Modbus can read live substation data through this interface. Register layout, data type conversion, and the failure modes specific to the Modbus-to-DNP3 translation.
5. **IEC 61850 Integration:** GOOSE messages carry protection relay trip and status signals across the station LAN in under four milliseconds. Sampled Values carry current and voltage waveforms. MMS handles operator control and data retrieval. How the RTAC maps each to its internal tag database.
6. **Local Logic:** When the WAN link to the SCADA master fails, the RTAC keeps running IEC 61131-3 programs locally. Event-driven automation, load shedding sequences, and protection coordination logic that must execute correctly with or without upstream communication. The logic engine, scan cycle behavior, and design principles for automation that survives communication loss.
7. **Security and Access Control:** Role-based access control limits which operators can view, control, or configure the RTAC. Serial port security prevents unauthorized local access. Firmware update procedures maintain integrity without interrupting the substation. NERC CIP implications for each configuration decision.
8. **Field Scenarios:** Three case studies from real substation automation work: a transformer bay with automatic tap changer response to load changes, a feeder protection scheme coordinating with downstream reclosers, and a load shedding sequence that must execute within 200 milliseconds of detecting an overload — with or without the SCADA master online.

Certification Relevance

- **SEL Certified Engineer** — SEL offers application engineer certification through their training center; RTAC configuration is a core track.
- **GICSP (GIAC)** — Substation automation, IEC 61850 security, and protocol gateway configuration appear in ICS security exam domains.
- **ISA CCST** — Protocol gateway integration and field device configuration tested at CCST Levels II and III.

Next Steps

1. Take DNP3 and IEC 61131-3 before RTAC. Protocol knowledge and programming skills are both prerequisites for field configuration work.
2. Download ACCELERATOR Architect (free from selinc.com) and configure a lab project against a virtual RTAC.
3. Attend SEL application engineering training in Pullman, WA or at a regional event. The in-person lab is the fastest path to real proficiency.
4. GICSP pairs well with RTAC expertise for engineers working in power utilities and substations.